(i) Find

$$\int x^2 e^x \, dx$$

**(4)**

(ii) Use the substitution $u = \sqrt{1 - 3x}$ to show that

$$\int \frac{27x}{\sqrt{1 - 3x}} \, dx = -2(1 - 3x)^{\frac{1}{2}}(Ax + B) + k$$

where $A$ and $B$ are integers to be found and $k$ is an arbitrary constant.

**(6)**